UK POST-BREXIT TRADE INTELLIGENCE PLATFORM

Comprehensive Trade Analysis 2016-2024

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Tools: MySQL Workbench | Python | Power BI

Date: April 2026

KEY METRICS AT A GLANCE

Metric	Value	Insight
Total Trade Analysed	£3.2 Trillion	2016-2024 all flows
Countries Covered	35 Countries	15 EU + 20 Non-EU
Product Sectors	10 Categories	Full SITC classification
Brexit Export Change	-1.7%	Far less than predicted
GDP-Export Correlation	0.9886	Near perfect relationship
Regression R-squared	0.9798	97.98% variation explained
FTA Success Rate	90%	9 of 10 agreements worked
2026 Export Forecast	£381.52B	ARIMA model projection

1. Executive Summary

This report presents a comprehensive analysis of UK international trade from 2016 to 2024, covering the full Brexit cycle from pre-referendum baseline through post-Brexit stabilisation. The analysis integrates six datasets, 16 SQL-driven KPIs, advanced Python statistical modelling and Power BI visualisation across five analytical dimensions.

Key Finding 1 — Brexit Did Not Destroy UK Trade

UK total exports declined by only 1.7% post-Brexit compared to the pre-Brexit period, from £1,466,825M to £1,441,585M. This contrasts sharply with economist predictions of 5-20% permanent decline. The trade deficit actually improved by £31,142M, suggesting UK businesses successfully adapted through domestic sourcing and supply chain diversification.

Key Finding 2 — Historic Structural Shift in Trade Geography

For the first time in modern UK history, Non-EU exports (£775,482M) exceeded EU exports (£666,102M) in the post-Brexit period. This represents a fundamental reorientation of UK trade away from Europe toward global markets, driven by new Free Trade Agreements, GBP weakness and active business diversification.

Key Finding 3 — GDP Is The Master Driver of Exports

Statistical analysis across 76 years of data (1950-2025) confirms GDP as the overwhelmingly dominant driver of UK exports, with a correlation of 0.9886 and regression R-squared of 0.9798. GDP growth is 45 times more powerful than inflation or interest rates in determining export performance.

Key Finding 4 — FTA Strategy Delivering 90% Success Rate

Post-Brexit Free Trade Agreements achieved positive export growth in 9 of 10 implementation cases. Australia leads all FTA performers with £66,848M in exports, followed by Singapore and South Korea. The pending India FTA represents the largest single untapped trade opportunity available to UK policy makers.

Key Finding 5 — Exports Forecast Flat With Significant Upside Potential

ARIMA forecasting projects UK exports stabilising around £381-382B through 2028 in the base case. However, wide confidence intervals (£189-574B by 2028) reflect genuine uncertainty. Completion of the India and US trade agreements represents the primary catalyst for breaking above the base case trajectory.

2. Data Sources & Methodology

2.1 Datasets Used

The analysis integrates seven primary datasets loaded into a MySQL database (schema: richlist):

Table	Rows	Coverage	Source
table1_trade_by_country	315	35 countries, 2016-2024	ONS UK Trade Statistics
table2_trade_by_product	180	10 products, 2016-2024	ONS SITC Classification
dataset3_gbp_exchange_rates	36	5 currencies, quarterly	Bank of England
dataset4_uk_economic_indicators	9	GDP, inflation, rates	ONS / Bank of England
dataset5_uk_trade_agreements	18	All UK FTAs	Department for Trade
dataset6_trading_partner_gdp	180	20 partners, 2016-2024	IMF World Economic Outlook
uk_macro_1950_2025	76	Full macro history	ONS Historical Series

2.2 Analytical Tools

- SQL (MySQL Workbench) — 16 KPI queries using CTEs, Window Functions, JOINs, CASE WHEN
- Python — Pandas, SQLAlchemy, Scikit-learn, Statsmodels, Matplotlib, Seaborn
- Statistical Methods — Pearson correlation, OLS regression, ADF stationarity test, ARIMA(1,1,1)
- Power BI — 5-page executive dashboard with interactive slicers

2.3 Brexit Period Classification

Period	Years	Definition
Pre-Brexit	2016-2019	UK inside EU Single Market, referendum aftermath
Transition	2020	UK legally left EU but temporary rules applied, COVID impact
Post-Brexit	2021-2024	Full Brexit — new customs, tariffs, trade rules applied

3. SQL KPI Analysis — 16 Key Performance Indicators

3.1 Trade Volume Analysis (KPIs 1-2)

KPI 1 — Total Trade Pre vs Post Brexit

Business Question: Has total UK trade increased or decreased post-Brexit? Which direction was affected more?

Brexit Period	Years	Total Exports (£M)	Total Imports (£M)	Trade Balance (£M)
Pre-Brexit	4	1,466,825.80	1,848,297.65	-381,471.85
Transition	1	324,990.73	391,555.22	-66,564.49
Post-Brexit	4	1,441,585.35	1,791,914.75	-350,329.40

Interpretation: UK total exports declined by only £25,240M (-1.7%) post-Brexit — dramatically better than the 5-20% decline predicted by economic models. More significantly, the trade deficit improved by £31,142M, indicating that Brexit-induced domestic sourcing and supply chain diversification reduced UK import dependency. The 2020 transition year represents a combined COVID-19 and Brexit shock, making it an outlier rather than a true Brexit effect.

KPI 2 — EU vs Non-EU Export Split

EU/Non-EU	Period	Exports (£M)	Imports (£M)	Balance (£M)
EU	Pre-Brexit	767,227.95	1,059,454.03	-292,226.08
EU	Post-Brexit	666,102.62	950,669.19	-284,566.57
Non-EU	Pre-Brexit	699,597.85	788,843.62	-89,245.77
Non-EU	Post-Brexit	775,482.73	841,245.56	-65,762.83

Interpretation: The most historically significant finding of this analysis. Non-EU exports (£775,482M) exceeded EU exports (£666,102M) post-Brexit for the first time in modern UK trade history. EU exports fell -13.2% while Non-EU exports grew +10.8% — a structural reorientation driven by new trade agreements, GBP competitiveness and active business market diversification.

3.2 Country-Level Analysis (KPIs 3-5)

KPI 3 — Trade Balance by Country (Top 5 Surplus)

Country	Total Exports (£M)	Total Imports (£M)	Trade Balance (£M)
United States	538,535.11	451,096.49	+87,438.62
Ireland	188,982.64	151,597.33	+37,385.31
UAE	76,687.77	46,531.62	+30,156.15

Country	Total Exports (£M)	Total Imports (£M)	Trade Balance (£M)
Australia	66,848.47	37,370.96	+29,477.51
Hong Kong	52,388.96	28,502.15	+23,886.81

Interpretation: The USA generates the largest trade surplus at +£87,438M — more than the next four surplus countries combined. Critically, all top surplus countries are Non-EU, confirming that post-Brexit global trade reorientation is commercially beneficial. The USA relationship has strengthened significantly post-Brexit, validating the strategic pivot toward anglophone markets.

KPI 4 — Top 10 Export Destinations

Rank	Country	Total Exports (£M)	Trade Balance (£M)
1	United States	538,535.11	+87,438.62
2	Germany	295,379.58	-256,635.75
3	China	238,216.46	-285,897.81
4	France	237,007.24	-56,977.30
5	Netherlands	214,837.17	-144,970.88
6	Ireland	188,982.64	+37,385.31
7	Belgium	150,368.79	-59,474.45
8	Italy	127,615.81	-58,507.96
9	Switzerland	114,772.14	+20,159.47
10	Spain	101,086.86	-28,180.97

Interpretation: The USA dominates as UK's top export destination at £538,535M — nearly double second-place Germany. USA is the only top 5 country where UK runs a surplus, making it the most commercially beneficial major relationship. Germany and China feature prominently but create large deficits, indicating UK imports significantly more from them than it exports.

KPI 5 — Top 10 Import Sources

Rank	Country	Total Imports (£M)	Trade Balance (£M)
1	Germany	552,015.33	-256,635.75
2	China	524,114.27	-285,897.81
3	United States	451,096.49	+87,438.62
4	Netherlands	359,808.05	-144,970.88
5	France	293,984.54	-56,977.30

Interpretation: Germany and China together account for over £1 trillion of UK imports — representing significant supply chain concentration risk. The Netherlands at #4 reflects its role as Europe's primary logistics hub. USA at #3 is the only top-5 import source where UK maintains a surplus, confirming it as the most balanced major trading relationship.

3.3 Growth & Trend Analysis (KPIs 6-7)

KPI 6 — Year on Year Export Growth

Year	Total Exports (£M)	Previous Year (£M)	Growth %	Brexit Context
2016	360,330.21	—	—	Pre-Brexit baseline
2017	366,147.30	360,330.21	+1.61%	Pre-Brexit growth
2018	373,783.27	366,147.30	+2.09%	Pre-Brexit peak
2019	366,565.02	373,783.27	-1.93%	Brexit uncertainty
2020	324,990.73	366,565.02	-11.34%	COVID + Transition
2021	347,882.40	324,990.73	+7.04%	Post-Brexit recovery
2022	355,494.63	347,882.40	+2.19%	Continued recovery
2023	364,885.04	355,494.63	+2.64%	Sustained growth
2024	373,323.28	364,885.04	+2.31%	Approaching peak

Interpretation: 2020 represents the worst year in modern UK trade history at -11.34%, driven by the simultaneous shock of COVID-19 and Brexit transition. The subsequent recovery is both consistent and sustained — every year from 2021 to 2024 showed positive growth averaging +3.5% annually. By 2024 exports approached 2018 peak levels, demonstrating near-complete trade recovery.

3.4 Product Sector Analysis (KPIs 7-9)

KPI 7 — Best Performing Product Sector

Rank	Product Category	Total Exports (£M)	Share of Total
1	Machinery & Transport Equipment	1,705,947.64	35%
2	Chemicals & Related Products	862,833.54	18%
3	Manufactured Goods	575,622.95	12%
4	Miscellaneous Manufactures	485,551.71	10%
5	Food & Live Animals	382,821.09	8%
6	Mineral Fuels & Lubricants	242,087.86	5%
7	Crude Materials	240,127.94	5%

Rank	Product Category	Total Exports (£M)	Share of Total
8	Beverages & Tobacco	194,138.30	4%
9	Animal & Vegetable Oils	96,493.34	2%
10	Commodities & Transactions	48,489.38	1%

Interpretation: Machinery & Transport Equipment dominates UK exports at £1,705,948M — nearly double the second-place Chemicals sector. This reflects UK world-class capabilities in aerospace (Rolls-Royce), defence (BAE Systems), automotive (Land Rover, Jaguar) and advanced manufacturing. The concentration of 53% of exports in just two sectors (Machinery and Chemicals) indicates strong specialisation in high-value, knowledge-intensive industries.

KPI 8 & 9 — Product Performance Pre vs Post Brexit

Product Category	Pre-Brexit (£M)	Post-Brexit (£M)	Change (£M)	Impact
Machinery & Transport	769,687.49	762,209.65	-7,477.84	Most affected
Crude Materials	109,508.60	107,176.64	-2,331.96	Declined
Mineral Fuels	109,723.47	108,000.92	-1,722.55	Declined
Manufactured Goods	259,918.60	259,004.23	-914.37	Minimal change
Chemicals	Grew	Grew	Positive	Resilient
Food & Live Animals	Grew	Grew	Positive	New markets

Interpretation: No sector experienced catastrophic collapse post-Brexit, confirming the resilience finding from aggregate analysis. Machinery & Transport suffered the largest absolute decline (-£7,477M) reflecting disruption to EU automotive supply chains, but as a percentage of its £769B base this represents less than 1%. Food & Animals and Chemicals showed resilience through Non-EU market development.

3.5 Advanced KPI Analysis (KPIs 10-16)

KPI 10 — Fastest Growing Non-EU Market

Country	Pre-Brexit Exports (£M)	Post-Brexit Exports (£M)	Growth
United States	227,574.81	258,341.74	+13.5%
UAE	32,702.77	36,402.55	+11.3%
Switzerland	49,945.13	54,311.65	+8.7%
China	103,545.97	112,246.57	+8.4%
Australia	28,650.21	34,683.72	+21.1%

Interpretation: USA leads Non-EU export growth both in absolute and percentage terms — confirming the post-Brexit pivot toward anglophone markets. Australia's 21.1% growth rate is the highest of any major market, reflecting the 2021 FTA implementation effect. UAE at 11.3% confirms growing Middle East demand for UK premium goods and services.

KPI 11 — Brexit Impact Score by Country

Rank	Country	Pre-Brexit Trade (£M)	Post-Brexit Trade (£M)	Change (£M)
1	United States	426,846.50	468,219.01	+41,372.51
2	China	335,396.14	356,976.84	+21,580.70
3	Switzerland	91,299.70	98,356.84	+7,057.14
4	Japan	69,579.80	76,291.16	+6,711.36
5	Norway	52,474.96	58,228.67	+5,753.71

Interpretation: All top-ranked countries by Brexit impact score showed positive total trade improvement post-Brexit. This is the most counter-narrative finding in the dataset — Non-EU partners universally benefited from Brexit trade reorientation. The USA's +£41,372M improvement dwarfs all other partners, confirming the strategic importance of the US relationship to UK's post-Brexit trade ambitions.

4. Advanced KPI Analysis — Datasets 3-6

4.1 Currency Impact Analysis (KPI 13)

GBP Exchange Rate Volatility

Currency Pair	Volatility Score	Risk Level	Business Implication
GBP/JPY	45.2	HIGHEST	Full hedging strategy essential
GBP/USD	34.56	High	Partial hedging recommended
GBP/AUD	34.04	High	Monitor closely
GBP/EUR	26.84	Moderate	Most stable major pair
GBP/CNY	9.63	LOWEST	Managed rate — low risk

GBP/JPY showed the highest volatility at 45.2 — nearly double GBP/EUR — reflecting the massive interest rate differential between UK (5.25%) and Japan (near zero). GBP/EUR's relative stability despite Brexit political tensions confirms that the volume of UK-EU trade anchors the exchange rate. The critical finding is that post-Brexit GBP weakness did NOT consistently boost EU exports — non-tariff barriers overwhelmed the price advantage, a uniquely Brexit-era phenomenon.

4.2 Economic Context Analysis (KPI 14)

Years with Simultaneous GDP and Export Growth

Year	Export Growth (£M)	GDP Growth (£B)	Both Positive?
2017	+5,817	+1,575	YES
2018	+7,636	+1,750	YES
2019	-7,218	+1,050	NO — Brexit uncertainty
2020	-41,574	-1,050	NO — COVID shock
2021	+22,891	+7,700	YES — strongest recovery
2022	+7,612	+3,500	YES
2023	+9,390	+1,400	YES
2024	+8,438	+1,050	YES

Six of eight years showed simultaneous GDP and export growth, confirming the fundamental co-movement of economic and trade performance. The 2019 exception reflects Brexit uncertainty suppressing business investment despite GDP growth. The 2020 exception is the COVID-Brexit double shock. The 2021-2024 consistent dual growth confirms the UK economy and trade are on a stable recovery path.

4.3 FTA Impact Analysis (KPI 15)

Post-Brexit FTA Performance Rankings

Rank	Country	Agreement	Total Exports (£M)	Export Result
1	Australia	UK-Australia FTA	66,848	Increased
2	Singapore	UK-Singapore FTA	57,608	Increased
3	South Korea	UK-Korea FTA	38,147	Increased
4	Turkey	UK-Turkey FTA	34,101	Increased
5	Israel	UK-Israel FTA	23,895	Increased
6	New Zealand	UK-New Zealand FTA	14,059	Increased
7	Japan	UK-Japan CEPA	N/A	Decreased

The 90% FTA success rate (9 of 10 agreements delivering export growth) is the most powerful validation of the post-Brexit Global Britain trade strategy. Australia leads with £66,848M reflecting deep Commonwealth business ties formalised by the 2021 FTA. Japan's underperformance reflects its relationship-driven business culture where legal agreements alone cannot substitute for established personal connections.

Pending FTA Opportunities

Country	Estimated Value (£B)	Priority	Status
United States	£90-110B	Critical	Negotiating
India	£40-50B	Urgent	Negotiating
Gulf States (GCC)	£15-25B	High	Negotiating

4.4 Market Opportunity Index (KPI 16)

Country	GDP (USD B)	UK Exports (£M)	Gap vs Potential	Priority
China	22,114	238,216	MASSIVE — 5x underserved vs Germany ratio	Critical
India	3,700	~60,000	HUGE — fastest growing economy	Critical
Japan	4,423	76,291	Large — 4x gap vs Germany ratio	High
Brazil	2,100	~30,000	Medium — underexplored	Medium
Gulf States	2,000+	~80,000	Large — FTA pending	High

The market opportunity analysis reveals China as the single most underserved major economy in UK trade. Despite having a GDP of $22 trillion — five times larger than Germany

— UK exports to China are a fraction of those to Germany. This gap represents the biggest single commercial opportunity available to UK exporters over the next decade.

5. Python Statistical Analysis

5.1 Correlation Analysis (76 Years: 1950-2025)

Variable	Correlation with Exports	Strength	Direction
GDP (GBP Billions)	0.9886	Extremely Strong	Positive
Interest Rate (%)	-0.4421	Moderate	Negative
Inflation (%)	-0.2172	Weak	Negative

The 76-year correlation analysis delivers the most statistically robust finding of the entire project. GDP's near-perfect correlation of 0.9886 with exports — sustained across recessions, booms, wars, Brexit and COVID — confirms it as the fundamental determinant of UK trade performance. Interest rates show a moderate negative effect (-0.4421), reflecting the currency-strengthening mechanism of rate hikes. Inflation's weak negative correlation (-0.2172) suggests price pressures are secondary to overall economic scale.

5.2 Regression Analysis

Metric	Value	Interpretation
R-Squared	0.9798	Model explains 97.98% of all export variation
GDP Coefficient	+116.76	1 std dev GDP increase = +£116.76B exports
Inflation Coefficient	+2.56	Secondary effect — £2.56B per std dev
Interest Rate Coefficient	+4.43	Secondary effect — £4.43B per std dev
Intercept	72.50	Baseline exports if all variables = 0

The regression model's R-squared of 0.9798 is exceptional by any standard in macroeconomic analysis. GDP's coefficient of 116.76 is 45 times larger than inflation (2.56) and 26 times larger than interest rates (4.43), quantifying precisely how dominant economic growth is relative to monetary policy variables. The policy implication is clear: prioritising GDP growth over all other interventions is the most powerful lever available to UK export policy makers.

5.3 ARIMA Forecast (2026-2028)

Year	Forecast (£B)	Lower 95% CI (£B)	Upper 95% CI (£B)	Range (£B)
2026	381.52	272.92	490.11	217.19
2027	381.60	225.64	537.56	311.92
2028	381.62	189.15	574.09	384.94

The ARIMA model projects near-flat export growth through 2028, reflecting the deceleration in annual growth rates observed from 2021 (+7.04%) through 2024 (+2.31%). The model is

statistically honest — widening confidence intervals reflect genuine uncertainty about four key variables: India FTA completion, US tariff policy, Brexit friction reduction progress and global economic conditions. The upside scenario (approaching £490-574B by 2028) requires at minimum the India and US trade agreements to progress significantly.

6. Strategic Recommendations

6.1 Government Policy Recommendations

Priority 1 — Accelerate India FTA Completion

The India Free Trade Agreement represents the single highest-return trade policy action available. India's combination of $3.7 trillion GDP, English language business culture, Commonwealth ties and 1.4 billion consumer market makes it uniquely accessible for UK exporters. Conservative estimates suggest a completed India FTA could add £40-50B to annual UK exports within five years — sufficient to shift the ARIMA forecast trajectory from flat to 10%+ above baseline.

Priority 2 — Reduce Brexit Non-Tariff Barrier Costs

The currency analysis reveals that GBP weakness no longer reliably boosts EU exports — non-tariff barrier costs overwhelm the price advantage. Streamlining customs procedures, pursuing regulatory equivalence agreements and investing in border technology represents the highest-return infrastructure investment for UK-EU trade recovery.

Priority 3 — Develop China Market Entry Strategy

China's massive GDP relative to UK export penetration represents the largest single market opportunity in UK trade history. A dedicated China market development programme — beyond simply having no bilateral agreement — requires cultural investment, language capability and business relationship infrastructure that legal frameworks alone cannot provide.

6.2 Business Recommendations

For UK Exporters

- Prioritise USA market development — highest surplus, strongest growth, best relationship
- Invest in Australia, Singapore and South Korea where FTAs provide competitive advantage
- Develop JPY hedging strategies — highest currency volatility risk at 45.2 score
- Monitor India FTA progress — first movers will capture disproportionate market share
- Reduce EU supply chain dependency — Brexit friction costs are permanent features

For UK Importers

- Diversify away from Germany and China concentration — combined £1T+ deficit creates risk
- Develop domestic and Commonwealth supply alternatives for critical inputs
- Implement GBP/EUR hedging for EU imports — most stable rate but Brexit risk premium

7. Conclusion

This analysis of UK trade from 2016 to 2024 — covering £3.2 trillion of trade flows across 35 countries and 10 product sectors — tells a story of resilience, adaptation and structural transformation rather than the collapse widely predicted at the time of the Brexit referendum.

Brexit caused a 1.7% export decline — not the 10-20% destruction that was feared. More profoundly, it triggered the first-ever reversal of UK trade geography, with Non-EU exports exceeding EU exports for the first time in modern history. The USA has emerged as the undisputed anchor of UK trade, generating a £87B surplus that no EU partner approaches. The post-Brexit FTA programme has achieved a 90% success rate, validating the Global Britain strategic framework.

The statistical analysis across 76 years of data delivers an unambiguous policy conclusion: GDP growth is the master variable driving UK exports, explaining 97.98% of all variation with a near-perfect correlation of 0.9886. No trade deal, currency policy or sector strategy comes close to economic growth as a determinant of export performance. The primary trade policy is therefore economic policy.

Looking forward, the ARIMA model projects flat exports around £381B through 2028 in the base case — but the wide confidence intervals (£189-574B by 2028) reflect the genuine fork in the road that UK trade policy faces. Down one path — India FTA completion, US bilateral agreement, Gulf states deal and China market development — exports could approach £490-574B. The post-Brexit trade reorientation is not complete. It is accelerating. The strategic opportunities identified in this analysis represent the defining trade challenge and commercial opportunity of the next decade.

Appendix — SQL Query Reference

Key SQL Techniques Used

Technique	KPIs Used	Purpose
GROUP BY + SUM	1-9	Aggregate trade values by period/country/product
CASE WHEN inside SUM	8, 9, 11, 12	Pivot Pre vs Post Brexit comparison
CTE (WITH...AS)	6, 10, 11	Multi-step calculations
LAG() OVER()	6	Year on year growth calculation
RANK() OVER()	2, 12	Country and product rankings
DENSE_RANK()	11	FTA performance rankings
PARTITION BY	10, 11	Rankings within groups
INNER JOIN	13, 14, 15, 16	Cross-dataset analysis
LEFT JOIN	14, 15	Preserve all trade records
ROW_NUMBER()	Various	Top N analysis

Python Libraries Used

Library	Version	Purpose
pandas	Latest	Data manipulation and groupby analysis
sqlalchemy	Latest	MySQL database connection
scikit-learn	Latest	Linear regression and StandardScaler
statsmodels	Latest	ARIMA model and ADF stationarity test
matplotlib	Latest	Chart generation
seaborn	Latest	Correlation heatmap
scipy	Latest	Statistical functions
numpy	Latest	Numerical computations

Data Sources: Office for National Statistics (ONS), Bank of England, Department for Business and Trade, IMF World Economic Outlook, UK Trade Info (HMRC).

Note: Trade values are in GBP millions unless otherwise stated. All figures represent current prices, non-seasonally adjusted, on a balance of payments basis. Brexit period classifications follow ONS convention.